🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.NOS.1

Division Expressions with Fractions and Whole Numbers

Lesson 6-1 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

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EVIDENCE LAB MISSION

The Fraction Vault

You are the lead forensic analyst at the Reyes Detective Agency. A 3-foot strip of evidence tape must be cut into exact 1/4-foot sections before it can be logged — and the vault that holds the case file only opens once you know how many pieces that makes. Interpret fraction division and crack the case.

My Notes
Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use models to divide a whole number by a fraction and a fraction by a whole number.

Language Objective: I can explain a division model using the words dividend, divisor, quotient, and how many fit.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Division of Fractions | Dividend | Divisor | Quotient | Reciprocal | Unit fraction

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. Dividing asks how many groups of one ________ fit inside another amount.
2. In 6 ÷ 1/2, the number 6 being divided is the ________.
3. In 6 ÷ 1/2, the number 1/2 we divide by is the ________.
4. The answer to a division problem is the ________.
5. The fraction you get by flipping the numerator and denominator is the ________.

6 A fraction with a numerator of 1, like 1/5, is a ________________ .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How does interpreting fraction division help the detective?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Division of Fractions** — Finding how many groups of one fraction fit inside another amount.
  División de fracciones — Hallar cuántos grupos de una fracción caben dentro de otra cantidad.
- ☐ **Dividend** — The number you are splitting up in a division problem.
  Dividendo — El número que estás repartiendo en una división.
- ☐ **Divisor** — The number you split by in a division problem.
  Divisor — El número entre el que divides en una división.
- ☐ **Quotient** — The answer when you divide.
  Cociente — La respuesta cuando divides.
- ☐ **Reciprocal** — A fraction turned upside down.
  Recíproco — Una fracción volteada de arriba abajo.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The detective can mark ____ sections because 6 ÷ 1/3 = ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: There will be more than 3 sections because each piece is smaller than a whole foot.

Evidence: Students reason that dividing by a number less than 1 gives a quotient greater than the dividend, so there are more than 3 sections.

Reasoning: This evidence connects the definition of division of fractions to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective can mark ____ sections because 6 ÷ 1/3 = ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Division of Fractions
2. **Notes 2:** Dividend
3. **Notes 3:** Divisor
4. **Notes 4:** Quotient
5. **Notes 5:** Reciprocal
6. **Notes 6:** Unit fraction
7. **Watch for this mistake:** *A common mistake in Interpret Division of Fractions is assuming that dividing always makes the answer smaller. For 4 ÷ 1/3, students reason 'dividing shrinks the number' and compute 4 ÷ 3 ≈ 1.3 instead of asking how many 1/3-foot pieces fit into 4 feet (each whole foot holds 3 thirds, so 4 × 3 = 12). Before you submit, check: if the divisor is a fraction smaller than 1, the quotient must be BIGGER than the dividend, not smaller.*
8. **Try It 1:** A. 5 ÷ 1/6 = 30 slivers — *How many 1/6-inch slivers fit into 5 inches? Each inch holds 6 sixths, so 5 × 6 = 30 slivers.*
9. **Try It 2:** — *3/4 ÷ 1/4 asks how many one-fourth pieces make up three-fourths. The fraction bar shows 3/4 already built out of fourth-size pieces, so counting those pieces gives the quotient: 3.*